

Korea Equity Strategy

Update on the de-leveraging process in Korea

While the fundamentals of the Korea market remain on solid ground (despite some concerns – see below), an **intense period of de-leveraging has driven equity prices lower**. The KOSPI index is now down a sizable -28% from the peak on 22 June in a highly volatile move. What was triggered by routine fundamental concerns and rotational flow, was amplified by leveraged ETFs, and in recent days increasingly has the hallmarks of hedge fund positioning unwind (especially given the -30% Price Momentum factor drawdown). The rapid pace of gains in the Korea market over the past year made it an inevitable destination of leverage - from retail investors, equity L/S hedge funds as well as macro funds - compounding these gains. As we have noted for the past 6-7 weeks now (see [here](#), [here](#)), some of the side effects of this growth have been the elevated volatility and forced foreign selling - producing a self-correcting mechanism. In the near term, our focus remains on the extent of normalization in these conditions (**we estimate leveraged ETF unwind to be 75% through and equity H/F de-leveraging >50% through**). Stepping back, though, our view remains that the fundamental strength in the market remains good for several years ahead (a combination of global AI spend, security/resilience spend, wealth effect across corporates/households/government and structural governance improvements), meaning that the **market should be able to overcome positioning unwinds and persevere with the positive trend. We remain OW Korea in our regional allocations and our 12m-out base case KOSPI target remains 12,500**.

- **Extent of leverage “normalization”:** Of all the leverage channels in Korea, we see (1) **futures & options** largely an institutional market with limited signs of speculative excess, and (2) **retail leverage via margin** and bank loans as relatively modest (down a lot in KOSDAQ though). The issues around (3) **swap capacity** (globally rising funding costs compounded by Korea-specific capital constraints and concentration issues) and (4) volatility from outstanding **leveraged ETFs** – both remain but have materially eased (tactically, because the market is lower, and structurally, because the underlying issues are being addressed). We estimate leveraged ETF unwind to be 75% through (to an acceptable level of ~\$18bn) and equity H/F de-leveraging more than 50% through (assuming L/S ratios reach the upper end of 2025’s range).
- **Leveraged ETFs:** Leveraged ETFs with Korea underlyings grew to \$50bn in AUM (mostly due to the growth of the market) by late June. Relative to its market size, this was 4x larger than the US and led to very pronounced volatility (and forced de-leveraging on down-days) - the VKOSPI to VIX ratio is still closer to 5x vs the usual 1x. Not only does this detract from vol-sensitive inflows, but it makes vol-based risk-management harder for asset managers and securities companies. Over Jun and July we have seen this excessive volatility compound the modest AI jitters seen elsewhere in global AI hardware stocks. Similar to previous deleveraging of leveraged products (Nikkei in 2015, Nasdaq in 2022, Oil in 2020, and most famously - even if somewhat differently - the inverse VIX products in 2018), this period of volatility has shrunk leveraged ETFs with Korea underlyings now to \$26bn. With new regulations aimed at keeping tighter checks on the growth of this space (minimum deposit requirement raised to KRW30mn from KRW10mn from

Equity Macro Research

Mixo Das ^{AC}

(852) 2800-0511

mixo.das@jpmorgan.com

J.P. Morgan Securities (Asia Pacific) Limited/ J.P. Morgan Broking (Hong Kong) Limited

Stanley Yang

(82-2) 758-5712

stanley.yang@jpmorgan.com

J.P. Morgan Securities (Far East) Limited, Seoul Branch

Rajiv Batra

(65) 6882-8151

rajiv.j.batra@jpmorgan.com

J.P. Morgan Securities Singapore Private Limited

Joy Wang

(852) 2800-2322

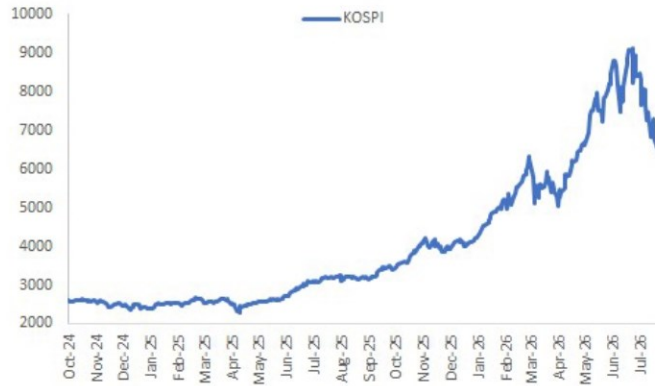
joy.y.wang@jpmorgan.com

J.P. Morgan Securities (Asia Pacific) Limited/ J.P. Morgan Broking (Hong Kong) Limited

Aug 5th, only cash will be recognized as the initial margin from Aug 19th, new listings of single-stock levered ETF products are temporarily halted, minimum trading unit for single-stock leveraged products will be raised from 1 share to 20 shares starting in Nov), further moderation in size is likely ahead.

- **HF leverage and swap capacity:** Swap capacity limits (for investors to access Korea equities via brokers) have been tightening for a while - faced with rapidly rising interest from global equity and macro investors (gross and net exposure towards Korea in the JPM Prime book have risen rapidly since April). This is partly a global story, where higher demand for financing (due to growth in markets and shift in AUM mix to higher leveraged entities) and tightening regulations having driven funding rates higher. It is also partly a Korea story, given the higher capital needed to match higher trading volumes (6-7x over the past year) and difficulty in managing concentration risks of investors unanimously preferring positions in the memory space. With the decline in market size and underperformance of memory names, though, this capacity tightness has eased substantially (a full normalization is still some distance away as brokers and investors work through these constraints). Similarly, in the present de-leveraging process, L/S ratios in the JPM Prime book have declined from highs of >5.5x to now <4x.
- **Retail leverage and margins:** Margin balances have declined somewhat from a high of >\$25bn to ~\$21bn now. Notably, most of the declines and margin calls have been in smaller-cap KOSDAQ market, and not in KOSPI. But at \$21bn, this is just 0.5% of equity market cap (vs 1.9% of market cap in the US and 2.8% in China A-shares). In addition, unlike leveraged ETFs that undergo a process of forced de-leveraging on any spot price declines, margin lending comes with buffers and discretion. Korean retail investors still have ample equity gains, cash balances, higher incomes and overseas assets to tap into to absorb any margin calls should they want to keep the position. We are thus less concerned about margin calls and de-leveraging through this channel becoming a systemic problem.
- **Record foreign outflows are slowing:** There has been >\$110bn of foreign outflows from Korea YTD - which would handily break the record of annual outflows from any market in Asia. The two memory names had become so large that they started to hit mandate limits for EM investors - which we estimated to be affecting roughly 10% of the foreign ownership in both stocks in 2Q. This forced investors to sell on rallies. Indeed, ~90% of YTD foreign outflows from Korea come from the memory names alone. This selling pressure has eased tactically as Korea (and particularly the memory names) has underperformed, meaning that the two heavyweights are now only 7.5% and 5.7% weights in MSCI EM (vs 9.5% and 8.3% in late June).
- **Fundamentals:** We remain [bullish on the AI cycle and related fundamentals](#). Earnings growth rate in Korea will slow from the scorching pace of 2026, but the **question in Korea now is less about growth and more about sustainability** (how much visibility can we have about higher earnings levels). While monetization at the model layer is being questioned again (with very positive reception to open-source models), at the hyperscaler layer the rental economics for data centers remain very positive. This means that hyperscalers should keep spending up. All our indicators on this are currently green across model capability, funding availability and rental rates. And external risks (like regulation, recession, etc.) don't look imminent. Memory demand has been questioned recently with reported technological and process breakthroughs reducing memory demand—but we are yet to see this in reality. And in terms of supply, the supply of memory equity should not be conflated with supply of physical memory. Outside of AI, there are earnings tailwinds from **growth in a variety of (AI-adjacent and other) Industrials**, potential **wealth-effect boost to Financials and Consumer**, and ongoing (if incremental) **valuation tailwinds from the governance reforms** (which will likely come back to prominence later this year).

Figure 1: KOSPI index has declined -29% from the peak of 22 Jun



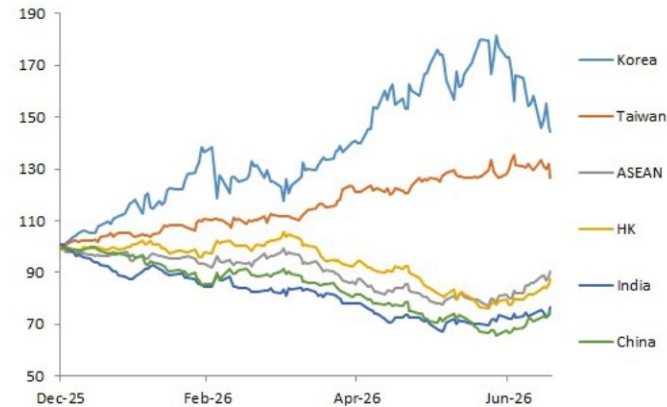
Source: Bloomberg Finance L.P., J.P. Morgan Equity Macro Research.

Figure 2: Bringing the index RSI to lowest levels since the start of this rally post Liberation Day in 2025



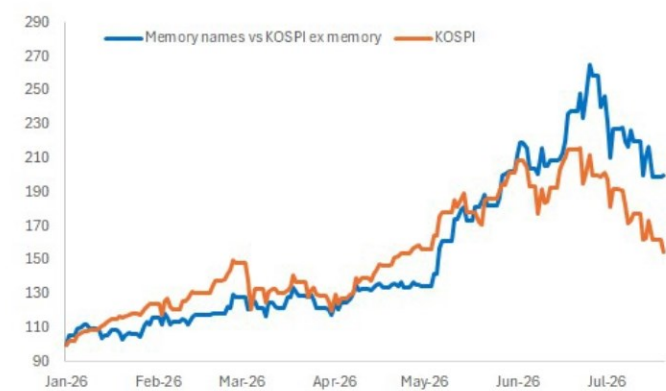
Source: Bloomberg Finance L.P., J.P. Morgan Equity Macro Research.

Figure 3: Korea is still the best performing market in the region YTD in USD terms



Source: Bloomberg Finance L.P., J.P. Morgan Equity Macro Research.

Figure 4: KOSPI has largely been a reflection of memory upside in recent months



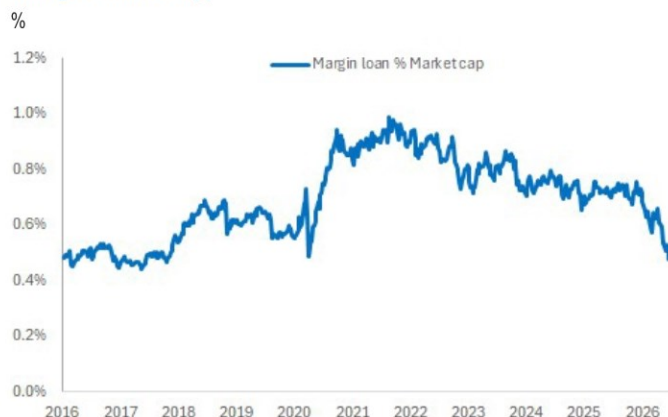
Source: Bloomberg Finance L.P., J.P. Morgan Equity Macro Research.

Figure 5: Margin balances and leveraged products are the primary channels for retail investors to access leveraged upside - in Korea margin balances are very acceptable, but leveraged ETFs outstanding are still large

\$bn	Market cap	Margin balances	% of Mcap	Leveraged ETFs outstanding	% of Mcap
US	80000	1502	1.9%	250	0.3%
Korea	4000	21	0.5%	26	0.7%
Japan	8800	43	0.5%	5	0.1%
China A	15000	422	2.8%	1	0.0%
Taiwan	5000	19	0.4%	8	0.2%

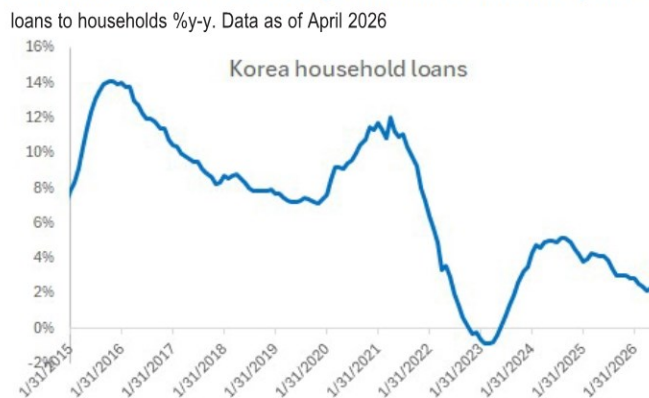
Source: Bloomberg Finance L.P., FINRA, KOFA, JPX, J.P. Morgan Equity Macro Research.

Figure 6: For retail investors, margin leverage is not too high (and falling vs market cap)



Source: Bloomberg Finance L.P., J.P. Morgan Equity Macro Research.

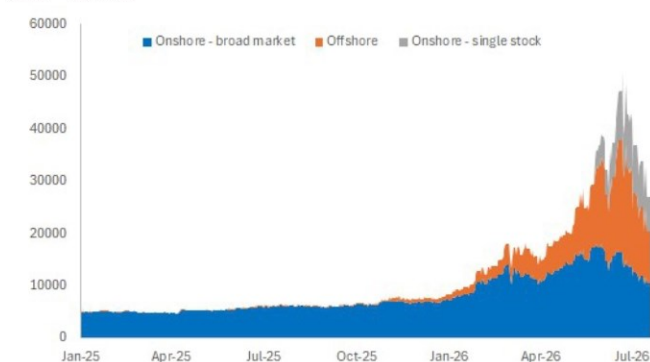
Figure 7: Similarly there is little evidence at the aggregate level that Korean households are borrowing from banks to invest in equities



Source: Bloomberg Finance L.P., J.P. Morgan Equity Macro Research.

Figure 8: In terms of leveraged ETFs, their AUM has declined from highs of \$50bn to now \$26bn

AUM - USD mn



Source: Bloomberg Finance L.P., J.P. Morgan Equity Macro Research.

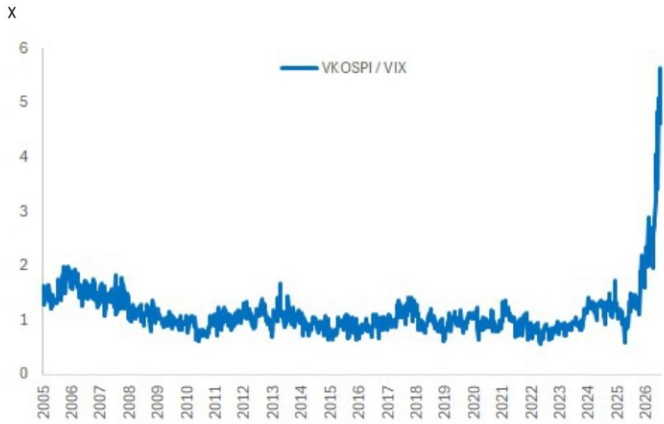
Figure 9: To be sure, much of this is simply due to the decline in the market, as flows have continued to actually be positive

cumulative inflows, USD mn



Source: Bloomberg Finance L.P., J.P. Morgan Equity Macro Research.

Figure 10: Largely as a result of leveraged ETF hedging, volatility in Korea has notably disconnected from volatility elsewhere - and yet to normalize



Source: Bloomberg Finance L.P., J.P. Morgan Equity Macro Research.

Figure 12: Separately, Korean equities have seen >\$110bn of outflows YTD, but most of this is due to mandate constraints for LOs on owning large-caps

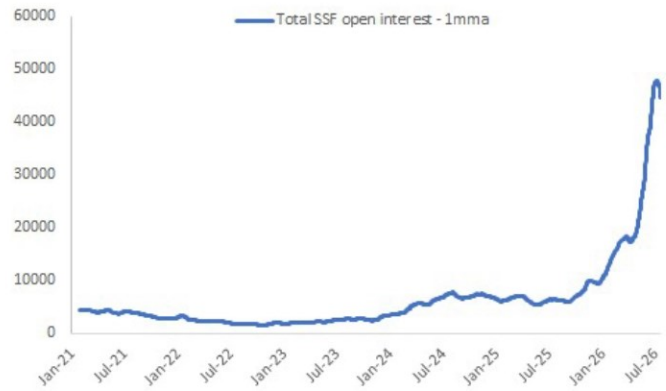
cumulative flow, \$mn, KRX data



Source: Bloomberg Finance L.P., J.P. Morgan Equity Macro Research.

Figure 11: The open interest of single stock futures (implementation channel of the leveraged ETFs) has started to decline now

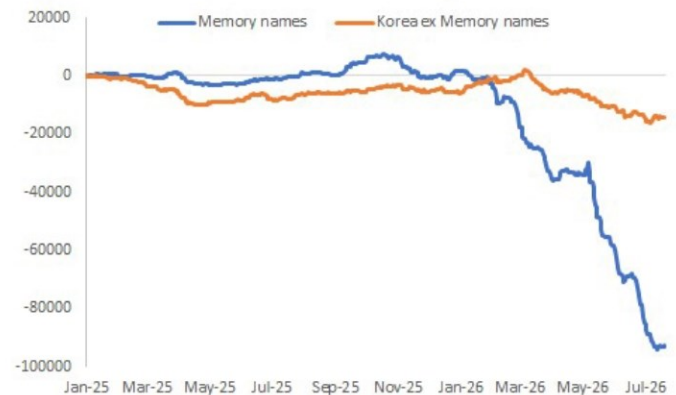
USD mn



Source: Bloomberg Finance L.P., J.P. Morgan Equity Macro Research.

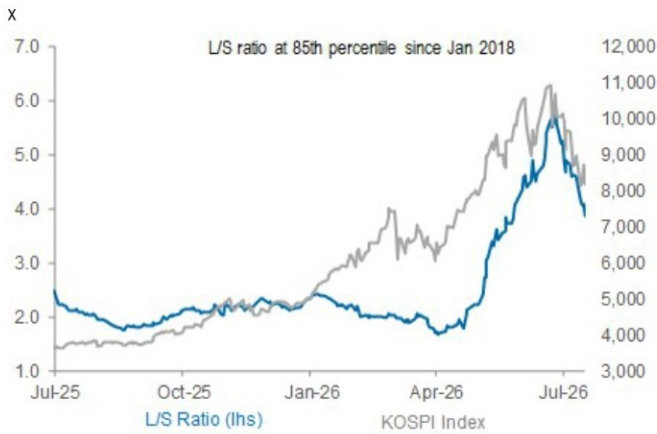
Figure 13: Indeed, 90% of the outflow is from the two memory names. The decline in EM index weights of these stocks has substantially slowed outflows

cumulative flow, \$mn



Source: Bloomberg Finance L.P., J.P. Morgan Equity Macro Research.

Figure 14: L/S ratios for hedge funds in the JPM Prime book has declined from highs, but not fully normalized



Source: Bloomberg Finance L.P., J.P. Morgan Positioning Intelligence.

Figure 16: Retail investors have been the main buyers of Korean equities - sentiment about equities and leverage is still strong

KRW bn, KRX data



Source: Bloomberg Finance L.P., J.P. Morgan Equity Macro Research.

Figure 18: Price momentum factor has seen a deep drawdown in recent weeks - similar to US, Japan etc

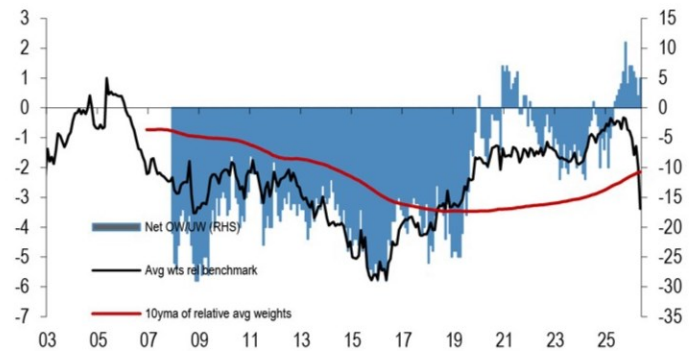
indicative performance of L/S quintiles

	Px Momentum	EPS revision	Growth	Quality	Value	Low Vol
Week ending 03-Jul	-10.8%	-4.9%	-0.8%	3.3%	0.9%	8.7%
Week ending 10-Jul	-7.9%	-3.1%	-2.1%	0.8%	5.6%	7.9%
Week ending 17-Jul	-6.3%	-3.0%	0.6%	1.9%	1.3%	7.0%
Week ending 24-Jul	-1.7%	0.4%	-1.0%	-0.2%	1.3%	3.4%

Source: Bloomberg Finance L.P., J.P. Morgan Equity Macro Research.

Figure 15: EM LO investors have become more UW Korea YTD as mandate constraints bite

weightings relative to the MSCI benchmark and net OW/UW



Source: Bloomberg Finance L.P., EPFR, MSCI, J.P. Morgan Equity Macro Research.

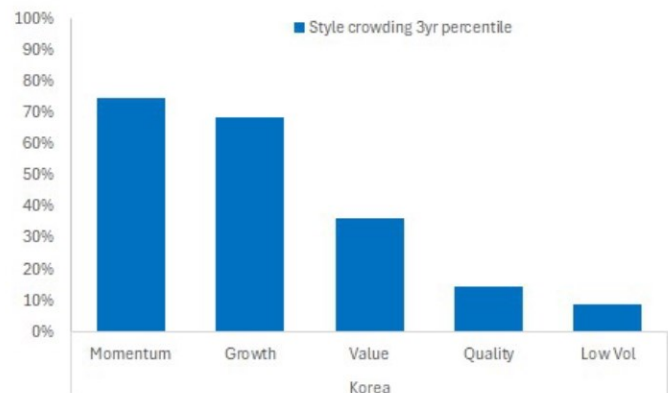
Figure 17: Since the beginning of June, most purchased overseas stocks by Korean retail investors include several leveraged products

\$bn



Source: Bloomberg Finance L.P., J.P. Morgan Equity Macro Research.

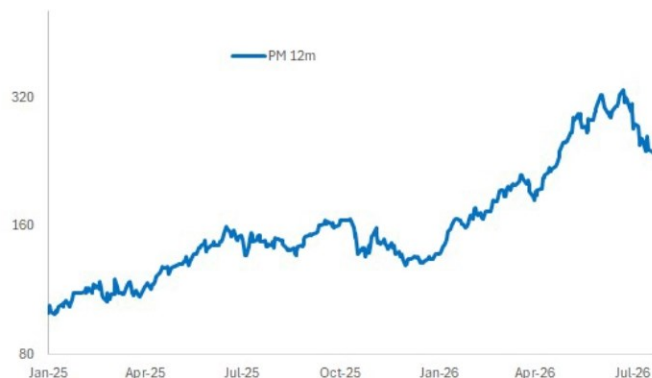
Figure 19: Momentum crowding is now starting to come off



Source: Bloomberg Finance L.P., J.P. Morgan Equity Macro Research.

Figure 20: Drawdown in Price Momentum factor still leaves strong YTD performance

indicative performance of L/S quintile - log scale



Source: Bloomberg Finance L.P., J.P. Morgan Equity Macro Research.

Figure 21: 3Q tends to be reasonably strong seasonally for Price Momentum factor performance in Korea - 4Q is more challenging

Average performance over past five years



Source: Bloomberg Finance L.P., J.P. Morgan Equity Macro Research.

Figure 22: Korea sector EPS revisions remain strong as earnings get underway

	2026 EPS			2027 EPS		
	1m	3m	6m	1m	3m	6m
Market	7.8%	35.5%	143.4%	12.2%	53.7%	199.8%
Tech	8.5%	42.2%	215.5%	13.1%	63.1%	303.3%
Financials	1.7%	5.7%	9.8%	2.3%	6.5%	11.1%
Industrials	11.8%	42.6%	91.0%	18.1%	59.4%	108.6%
Discretionary	0.5%	-4.5%	-8.9%	0.8%	-3.3%	-6.8%
Materials	-5.8%	-9.0%	-13.9%	-2.3%	-1.2%	1.4%
Staples	3.0%	7.5%	11.7%	3.2%	8.2%	12.8%
Healthcare	0.6%	-3.2%	1.6%	0.9%	-6.3%	-3.0%
Comm Serv	-0.9%	-5.4%	-10.0%	-0.2%	0.1%	-4.5%
Energy	-4.5%	92.5%	127.7%	1.0%	34.2%	52.6%

Source: Refinitiv, J.P. Morgan Equity Macro Research.

Analyst Certification: The Research Analyst(s) denoted by an “AC” on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an “AC” on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst’s personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst’s compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst’s analysis was made in good faith and that the views reflect the Research Analyst’s own opinion, without undue influence or intervention.

All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Important Disclosures

Company-Specific Disclosures: J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan–covered companies, and certain non-covered companies, by visiting <https://www.jpmm.com/research/disclosures>, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

Explanation of Equity Research Ratings, Designations and Analyst(s) Coverage Universe:

J.P. Morgan uses the following rating system: Overweight (over the duration of the price target indicated in this report, we expect this stock will outperform the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe); Neutral (over the duration of the price target indicated in this report, we expect this stock will perform in line with the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe); and Underweight (over the duration of the price target indicated in this report, we expect this stock will underperform the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe. NR is Not Rated. In this case, J.P. Morgan has removed the rating and, if applicable, the price target, for this stock because of either a lack of a sufficient fundamental basis or for legal, regulatory or policy reasons. The previous rating and, if applicable, the price target, no longer should be relied upon. An NR designation is not a recommendation or a rating. Some stocks under coverage have a rating but no price target; in these cases, we expect the stock will outperform/perform in line/underperform the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe of the relevant duration of the region. In our Asia (ex-Australia and ex-India) and U.K. small- and mid-cap Equity Research, each stock’s expected total return is compared to the expected total return of a benchmark country market index, not to those Research Analysts’ coverage universe. If it does not appear in the Important Disclosures section of this report, the certifying Research Analyst’s coverage universe can be found on J.P. Morgan’s Research website, <https://www.jpmorganmarkets.com>.

J.P. Morgan Equity Research Ratings Distribution, as of July 04, 2026

	Overweight (buy)	Neutral (hold)	Underweight (sell)
J.P. Morgan Global Equity Research Coverage*	53%	36%	12%
IB clients**	83%	80%	73%
JPMS Equity Research Coverage*	51%	37%	12%
IB clients**	95%	92%	87%

*Please note that the percentages may not add to 100% because of rounding.

**Percentage of subject companies within each of the "buy," "hold" and "sell" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.

For purposes of FINRA ratings distribution rules only, our Overweight rating falls into a buy rating category; our Neutral rating falls into a hold rating category; and our Underweight rating falls into a sell rating category. Please note that stocks with an NR designation are not included in the table above. This information is current as of the end of the most recent calendar quarter.

Equity Valuation and Risks: For valuation methodology and risks associated with covered companies or price targets for covered companies, please see the most recent company-specific research report at <http://www.jpmorganmarkets.com>, contact the primary analyst or your J.P. Morgan representative, or email research.disclosure.inquiries@jpmorgan.com. For material information about the proprietary models used, please see the Summary of Financials in company-specific research reports and the Company Tearsheets, which are available to download on the company pages of our client website, <http://www.jpmorganmarkets.com>. This report also sets out within it the material underlying assumptions used.

History of Investment Recommendations:

A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of <http://www.jpmorganmarkets.com> where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Registration of non-US Analysts: Unless otherwise noted, the non-US analysts listed on the front of this report are employees of non-US affiliates of J.P. Morgan Securities LLC, may not be registered as research analysts under FINRA rules, may not be associated persons of J.P. Morgan Securities LLC, and may not be subject to FINRA Rule 2241 or 2242 restrictions on communications with covered companies, public appearances, and trading securities held by a research analyst account.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to [UK MIFID Research Unbundling exemption](#) for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

All research material made available to clients are simultaneously available on our client website, J.P. Morgan Markets, unless specifically permitted by relevant laws. Not all research content is redistributed, e-mailed or made available to third-party aggregators. For all research material available on a particular stock, please contact your sales representative.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Any digital or crypto assets discussed in this research report are subject to a rapidly changing regulatory landscape. For relevant regulatory advisories on crypto assets, including bitcoin and ether, please see <https://www.jpmorgan.com/disclosures/cryptoasset-disclosure>.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC ("JPMS") acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit <https://www.theocc.com/components/docs/riskstoc.pdf> for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries ("J.P. Morgan Private Bank"), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst's name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst's name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:

Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina ("BCRA"- Central Bank of Argentina) and Comisión Nacional de Valores ("CNV"- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited ("JPMSAL") (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting <https://www.jpmm.com/research/disclosures>. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia's Research Independence Policy which can be found at the following link: [J.P. Morgan Australia - Research Independence Policy](#).

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissão de Valores Mobiliários (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P.Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Superintendencia Financiera de Colombia (SFC). Any reference in this material to products or services offered abroad by entities other than the Bank in Colombia is included exclusively for descriptive purposes. Such references do not constitute, and should not be construed as, promotional activity or the provision of financial products or services within Colombian territory, as defined under applicable Colombian regulation.

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE ("JPM SE"), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions ("EEA professional investors"). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai - 400098, is registered with the Securities and Exchange Board of India (SEBI) as a 'Research Analyst' having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number - INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: <http://www.jpmpi.com>. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Ashutosh Sharma; ashutosh.j.sharma@jpmchase.com; +912261575002. Grievance Officer: Ramprasad K, jpmpi.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit [Terms and Conditions and Most Important Terms and Conditions \(MITC\)](#). The annual Compliance audit report is available at <http://www.jpmpi.com/#research>.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V., J.P. Morgan Grupo Financiero is member of the Mexican Stock Exchange ("Bolsa Mexicana de Valores") and the Institutional Stock Exchange ("Bolsa Institucional de Valores"), and it is authorized to act as a broker dealer by the National Banking and Securities Exchange Commission ("Comisión Nacional Bancaria y de Valores").

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMS) [MDDI (P) 057/08/2025 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange ("Non-Taiwan Listed Securities"), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities.** According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important Disclosures" in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission. The registered address is 548 One City Center Building, 50th Floor, Ploenchit Road, Lumpini, Pathum Wan, Bangkok 10330.

UK: Research is produced in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited ("JPML Ltd") which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA's policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: [J.P. Morgan EMEA - Research Independence Policy](#).

U.S.: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: <https://www.jpmorgan.com/disclosures/email>

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to [msci.com/disclaimer](https://www.msci.com/disclaimer)

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at <https://www.sustainalytics.com/legal-disclaimers>. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised July 04, 2026.

Copyright 2026 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any third-party artificial intelligence ("AI") systems or models when such J.P. Morgan Data is accessible by a third-party.

Completed 21 Jul 2026 03:22 AM HKT

Disseminated 21 Jul 2026 03:22 AM HKT